GLOBAL SECTIONS OF A SHEAF ON THE ETALE SITE OF A FIELD ARE THE GALOIS INVARIANTS OF ITS STALK

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument and even offers a second, direct proof; nothing but the identification of the last step of the formal chain as clear is left to the reader.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Equivalence between sheaves of sets and Galois sets. Same chapter, Theorem theorem-equivalence-sheaves-point. Turns the given sheaf into a G-set whose invariants compute the global sections.
- Sheaves represented by objects of the site. Sites, Sections sites-section-presheaves and sites-section-representable-sheaves. Identifies the global sections with morphisms from the sheaf represented by the site's final object.

Source reference. Original section 00V1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8690–8690. Navigation pointer invoked at proof line 8690; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 00WO: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8691–8691. Navigation pointer invoked at proof line 8691; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original proposition 03PC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2592–2592. Navigation pointer invoked at proof line

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8705; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Theorem 0.1. *Let $S = \mathrm{Spec}(K)$ with K a field. Let $\overline{s}$ be a geometric point of S. Let $G = Gal_{\kappa(s)}$ denote the absolute Galois group. Taking stalks induces an equivalence of categories*

$$Sh(S_{étale}) \longrightarrow G\text{-}Sets, \quad \mathcal{F} \longmapsto \mathcal{F}_{\overline{s}}.$$

Lemma 0.2. *Assumptions and notations as in Theorem 0.1. There is a functorial bijection*

$$\Gamma(S, \mathcal{F}) = (\mathcal{F}_{\overline{s}})^G$$

Proof. We can prove this using formal arguments and the result of Theorem 0.1 as follows. Given a sheaf $\mathcal{F}$ corresponding to the G-set $M = \mathcal{F}_{\overline{s}}$ we have

$$\begin{aligned} \Gamma(S, \mathcal{F}) &= \mathrm{Mor}_{Sh(S_{étale})}(h_{\mathrm{Spec}(K)}, \mathcal{F}) \\ &= \mathrm{Mor}_{G\text{-}Sets}(\{*\}, M) \\ &= M^G \end{aligned}$$

Here the first identification is explained in Sites, Sections 00V1 and 00WO, the second results from Theorem 0.1 and the third is clear. We will also give a direct proof[1].

Suppose that $t \in \Gamma(S, \mathcal{F})$ is a global section. Then the triple $(S, \overline{s}, t)$ defines an element of $\mathcal{F}_{\overline{s}}$ which is clearly invariant under the action of G. Conversely, suppose that $(U, \overline{u}, t)$ defines an element of $\mathcal{F}_{\overline{s}}$ which is invariant. Then we may shrink U and assume $U = \mathrm{Spec}(L)$ for some finite separable field extension of K, see Proposition 03PC. In this case the map $\mathcal{F}(U) \to \mathcal{F}_{\overline{s}}$ is injective, because for any morphism of étale neighbourhoods $(U', \overline{u}') \to (U, \overline{u})$ the restriction map $\mathcal{F}(U) \to \mathcal{F}(U')$ is injective since $U' \to U$ is a covering of $S_{étale}$. After enlarging L a bit we may assume $K \subset L$ is a finite Galois extension. At this point we use that

$$\mathrm{Spec}(L) \times_{\mathrm{Spec}(K)} \mathrm{Spec}(L) = \coprod\nolimits_{\sigma \in \mathrm{Gal}(L/K)} \mathrm{Spec}(L)$$

where the maps $\mathrm{Spec}(L) \to \mathrm{Spec}(L \otimes_K L)$ come from the ring maps $a \otimes b \mapsto a\sigma(b)$. Hence we see that the condition that $(U, \overline{u}, t)$ is invariant under all of G implies that $t \in \mathcal{F}(\mathrm{Spec}(L))$ maps to the same element of $\mathcal{F}(\mathrm{Spec}(L) \times_{\mathrm{Spec}(K)} \mathrm{Spec}(L))$ via restriction by either projection (this uses the injectivity mentioned above; details omitted). Hence the sheaf condition of $\mathcal{F}$ for the étale covering $\{\mathrm{Spec}(L) \to \mathrm{Spec}(K)\}$ kicks in and we conclude that t comes from a unique section of $\mathcal{F}$ over $\mathrm{Spec}(K)$. □

[1] For the doubting Thomases out there.